Federated O-RAN Slice SLA Prediction: A Cross-Architecture Empirical Benchmark on Colosseum/ColO-RAN

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***Abstract*—Federated learning (FL) on O-RAN edge gNBs is the natural paradigm for slice-level SLA-violation forecasting because per-user telemetry cannot be centrally pooled. Despite a decade of FL benchmark research, the deployment design space — sequence model architecture, federation algorithm, client-heterogeneity partition, commodity edge hardware — has been studied one axis at a time, leaving the joint behaviour under realistic O-RAN traffic characterised only anecdotally. This paper presents the first cross-architecture empirical FL benchmark on the publicly-available Colosseum/ColO-RAN testbed: 3 sequence backbones (LSTM, Mamba, Spiking-SSM) $\times$ 5 algorithms (FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn) $\times$ 6 partitions (natural-by-BS + Dirichlet $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$) $\times$ 10 seeds $=$ 900 cells, with NVML idle-baseline-subtracted training-energy measurement on RTX 4080 commodity hardware, augmented by a 15-cell controlled random-split ablation on $4\times$ Tesla V100-SXM2-32GB to probe the leading mechanism hypothesis. We report four empirical findings: (1) client heterogeneity in O-RAN is structurally helpful, not harmful — the natural base-station partition uniformly outperforms every parametric Dirichlet α across all 3 architectures $\times$ 5 algorithms, inverting the standard FL-benchmark assumption; (2) the algorithm-design space is flat at FedAdam-saturated headroom ($\leq +0.016$ AUC over FedAvg), bounding the room available to recent sharpness-aware methods; (3) selective-state-space backbones interact catastrophically with control-variate algorithms at moderate partition skew (Mamba $\times$ SCAFFOLD $\times$ $\alpha \in \{0.10, 0.50\}$); (4) architecture choice traverses $10\times$ in commodity-GPU energy and an order-of-magnitude larger AUC range than algorithm choice, dwarfing the algorithmic-leverage scale. We provide a reference benchmark with paired-bootstrap CI_{95} over 270 paired distributions, NVML-instrumented per-cell energy, and pre-registered statistical protocol; the artefact is released under Apache-2.0 with Croissant dataset metadata and a Zenodo DOI on paper acceptance.**

***Index Terms*—Federated learning, O-RAN, slice management, SLA prediction, state-space models, spiking neural networks, NVML energy measurement, paired-bootstrap inference.**

I. Introduction

OPEN Radio Access Networks (O-RAN) decomposes the cellular base station into commodity hardware components controlled by ML-driven xApps and rApps running on disaggregated RAN Intelligent Controllers (RICs) [1]. Slice-level Service Level Agreement (SLA) violation forecasting is one near-RT RIC xApp pattern of practical interest: each gNB simultaneously serves heterogeneous traffic slices (the ColO-RAN release exercises eMBB constant-bitrate, MTC Poisson, and URLLC Poisson — see Section **??**), and proactive forecasting of next-second SLA risk informs near-RT RIC resource-allocation decisions within its 10 ms – 1 s control loop. Because edge gNBs accumulate per-user telemetry that cannot be centrally pooled, federated learning (FL) is the natural training paradigm: each gNB trains a local sequence model on its own traces and shares only weight updates with the FL aggregator. We place the aggregator in the non-RT RIC / SMO orchestration layer (≥ 1 s timescale fits our 100-round federated training cadence per Section **??**); a near-RT-RIC-internal xApp aggregator would alternatively be possible if cross-cell xApp messaging supports the federation protocol — we do not engage with that deployment alternative beyond noting it.

Standard FL practice — codified by a decade of toy-image benchmarks — treats client heterogeneity as a *wound* to be healed: lower Dirichlet α (more skew) drives lower accuracy, motivating sharpness-aware methods, control variates, and proximal regularizers. **In O-RAN slice SLA prediction, this premise inverts.** When clients are the *natural* base stations of a Colosseum/ColO-RAN testbed (one client per gNB), the federated average across 7 base-station-natural clients consistently outperforms every uniform-Dirichlet partition of the same training data on the same model + algorithm. Empirically (Section **??**, all values are out-of-sample test AUC averaged over 10 seeds), the natural-by-BS partition achieves AUC 0.913–0.918 on LSTM (5-algo range; $\Delta \approx 0.005$), 0.8998–0.9186 on Mamba (with SCAFFOLD as the lower outlier and the other four algorithms in 0.911–0.919), and 0.840–0.856 on Spiking-SSM, **strictly above the AUC obtained at any parametric Dirichlet** α for every (arch, algo) pair, and monotonically increasing as $\alpha \to 0$ within the Dirichlet sweep. The $\alpha = 5.0 \to \alpha = 0.05$ AUC gap is ≈ 0.10–0.12 on the dense LSTM/Mamba backbones; on Spiking-SSM the gap is smaller (≈ 0.022) but still positive, and across all five algorithms heterogeneity is associated with higher AUC, not lower. The mechanism is open — natural-by-BS is *not* the low-α limit of the Dirichlet sweep (the two partition families redistribute rows along orthogonal axes, see Section **??**) — and we present a controlled `random_split` ablation in Section **??** (15 cells $\times$ 3 architectures on $4\times$ Tesla V100-SXM2-32GB) showing that any partition that breaks bs grouping (whether by per-slice Dirichlet redistribution at $\alpha = 5.00$ or by uniform random shuffling) collapses AUC to a tight band ~ 0.18 below natural-by-BS, and is statistically equivalent to Dirichlet $\alpha = 5.00$ within seed σ. The mechanism direction (preservation of bs

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grouping) is therefore confirmed; the precise dataset axis the model exploits via bs grouping (channel-state features, see Section **??**) remains under further investigation.

Despite the explosion of FL benchmarks since 2018, the deployment design space for FL-based slice SLA prediction remains under-characterized. The space spans four axes:

1) **Sequence model architecture** — recurrent (LSTM), structured state-space (Mamba), or bio-inspired sparse spiking variants (Spiking-SSM).
2) **Federation algorithm** — FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn (and the more recent sharpness-aware family discussed in Section II).
3) **Client heterogeneity** — including the natural base-station partition (one gNB $\Rightarrow$ one client) and parametric Dirichlet stress $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$ over slice IDs.
4) **Commodity hardware** — consumer-tier edge GPUs (RTX-class) at the gNB rather than data-center accelerators.

Existing benchmarks address each axis in isolation. NeurIPS-class FL benchmarks evaluate algorithm $\times$ heterogeneity on toy image data [2], [3], [4], [5]. Spiking-NN benchmarks compare architectures in centralized regimes on neuromorphic-friendly datasets [6]. FL $\times$ wireless papers focus on a single architecture [7], [8], [9]. The implicit assumption underlying single-axis recommendations: **best architecture $\times$ best algorithm $\times$ representative α $\times$ commodity hardware $=$ best deployment**.

This paper tests the assumption empirically. We construct the first comprehensive 4-axis FL benchmark on the publicly-available Colosseum/ColO-RAN testbed dataset [1], [10], sweeping {LSTM, Mamba, Spiking-SSM} $\times$ {FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn} $\times$ {natural-by-BS, Dirichlet $\alpha \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$} $\times$ 10 seeds $=$ 900 cells (90 groups, all 10 seeds populated), with NVML-instrumented per-round GPU energy measurement on RTX 4080 (sm_89) commodity hardware. Statistical inference uses paired-bootstrap CI_{95} on per-seed AUC deltas; Wilcoxon p-values are reported uncorrected per finding (with family-wise Bonferroni discussed in Section **??** as a supplementary check).

Contributions:

1) **Inverted heterogeneity in O-RAN slice federation**: against the standard FL-benchmark assumption, the natural base-station partition uniformly outperforms every parametric Dirichlet α in our sweep across all 3 architectures and 5 algorithms. We do not claim a closed-form mechanism — Section **??** reports a controlled `random_split` ablation that tests the leading "bs-conditioned signal preservation" hypothesis — but the empirical pattern is the first published characterization on Colosseum/ColO-RAN and is deployment-relevant regardless of which mechanism candidate ultimately survives.
2) **Algorithm-design space is flat in this regime; FedAdam achieves the empirical maximum in our 5-algorithm baseline**: across Dirichlet $\alpha \in \{0.05..5.0\}$, the spread between the best and worst of {FedAvg, FedProx, FedAdam, SCAFFOLD, FedDyn} on a fixed (arch, partition) cell is 0.022–0.031 AUC on LSTM/Spiking and 0.078 on Mamba (driven by a single SCAFFOLD outlier). FedAdam consistently leads FedAvg via paired-bootstrap CI_{95} (LSTM Dirichlet-averaged $\Delta = +0.0080$, range $+0.0048.. + 0.0128$, all 5 cells significant; Mamba Dirichlet-averaged $\Delta = +0.0062$, range $-0.0021.. + 0.0097$, 4 of 5 cells significant; Spiking Dirichlet-averaged $\Delta = +0.0164$, range $+0.0108.. + 0.0267$, 4 of 5 cells significant). The mechanism analysis in Sections II-F + **??** + **??** argues that LookAhead-style server-side aggregation (FedSWA / FedSCAM) is *unlikely to systematically exceed* this empirical maximum because of the variance-estimation gap between Adam-style and LookAhead-style server steps (predictive claim, not theorem); we accordingly dismiss FedSWA [11] and FedSCAM [12] via mechanism analysis rather than benchmark them. We do not benchmark FedBN [13] either; Section **??** L2 discusses why the algorithm-flatness ranking is unlikely to be overturned by FedBN given its reported gain envelope on related cellular feature-skew tasks.
3) **Architecture $\times$ algorithm catastrophic interaction (Mamba $\times$ SCAFFOLD $\times$ $\alpha \in \{0.10, 0.50\}$)**: SCAFFOLD on Mamba at $\alpha = 0.10$ yields AUC 0.7609 ± 0.0830 — std amplified $\sim 3\times$ over (Mamba, FedAvg) at the same α (0.0251) and $\sim 10\times$ over (Mamba, FedAvg) at $\alpha = 5.0$ (0.0073). The deployment warning is concrete: control-variate methods interact pathologically with selective-scan SSMs under high heterogeneity, even as both components individually behave well on neighboring cells.
4) **Architecture leverage dominates algorithm leverage** (paired-bootstrap CI_{95}) **on RTX 4080**: on the energy-AUC Pareto front (Section **??**), architecture choice spans $10\times$ in NVML-measured model-attributable energy on RTX 4080 (LSTM $\sim 3.5\,\mathrm{kJ} \rightarrow$ Mamba $\sim 10.5\,\mathrm{kJ} \rightarrow$ Spiking $\sim 41\,\mathrm{kJ}$ per cell). Holding (algorithm, IID partition) fixed, the paired Spiking $-$ LSTM AUC delta is concentrated at -0.061 to -0.074 across all 5 algorithms (all 5 cells significant at $p < 0.005$), substantively dwarfing the FedAdam $-$ FedAvg algorithmic gap of $+0.006$ to $+0.016$ on the same architectures (contribution 2). The $10\times$ span is hardware-specific: the same architectures on a sparsity-aware accelerator (Loihi-2, IBM NorthPole) would yield a substantially smaller LSTM-vs-Spiking energy ratio (Section **??** + Section **??** L1, L6). Operator decisions on commodity edge GPU should center architecture-energy trade-off, not algorithm breadth.
5) **Open reproducible reference benchmark**: spec-driven YAML pipeline with pre-flight algorithm validation and `--skip-completed` resumability; paired-bootstrap statistical pipeline; NVML idle-baseline-subtracted training energy; Croissant metadata; Zenodo DOI on acceptance; Apache-2.0 license (preregistered). The full 4-axis table maps deployment configurations $\rightarrow$ (AUC, F1, energy,

paired CI_{95}) over 90 (arch $\times$ algorithm $\times$ partition) groups, enabling lookup-style operator decisions on Colosseum/ColO-RAN.

The remainder of the paper is organized as follows. Section II reviews related work across the four axes. Section **??** describes the Colosseum/ColO-RAN dataset and our preprocessing pipeline. Section **??** details the architecture, algorithm, and partition methodology. Section **??** enumerates the reproducibility infrastructure and our Croissant-aligned data release. Section **??** reports results across the 90 (arch $\times$ algorithm $\times$ partition) groups with paired-bootstrap CI_{95}. Section **??** discusses the mechanism finding and articulates the applicability boundary on commodity edge hardware. Section **??** enumerates limitations. Section **??** concludes.

II. Related Work

A. Federated learning benchmarks

The canonical FL benchmark is **LEAF** [2], which standardized federated MNIST/Shakespeare evaluation pipelines for early FedAvg variants. **Flower** [14] later provided a framework rather than a benchmark, supporting heterogeneous device experiments. **FedScale** [3] scaled to thousands of cross-device clients; **FLAIR** [4] introduced a long-tail multi-label image dataset. **pfl-research** [5] integrates differential-privacy mechanisms into a simulation framework. Most recently, **fev-bench** [15] introduced rigorous bootstrap-CI_{95} evaluation for time-series forecasting, though without the federated dimension. None of these benchmarks targets O-RAN slice SLA prediction specifically; their toy-data settings cannot capture the covariate-skew structure of real RAN telemetry.

B. FL on cellular and wireless networks

A first generation of FL $\times$ cellular work targeted SLA-constrained slicing under the Beyond-5G banner. **Statistical FL for B5G SLA-Constrained RAN Slicing** [16] introduced a long-term CDF SLA constraint. **CDF-Aware FL** [17] formalized the constraint as a per-client penalty. **Fed-LSTM** [7] deployed an LSTM forecaster for slice-level traffic. **TRACTOR** [18] integrated reinforcement-learning xApps for resource-block assignment. **REAL** [19] integrated the OSC near-RT RIC with srsRAN for RL-based slice-resource-allocation xApps under urban-mobility channel modelling. Recently **Hayek et al. 2025** [20] measured FL convergence on a 5G/WiFi/Ethernet COTS testbed with Raspberry-Pi clients, exposing uplink straggler effects. **arXiv:2508.08479** [8] benchmarked FedAvg/FedProx/FedBN on five throughput-prediction datasets with LSTM, CNN, CNN+LSTM, and Transformer architectures, finding FedBN superior under feature skew. **Mangi et al. 2026** [9] ("When Clients Drift") proposed regime-aware aggregation under leave-one-regime-out evaluation on synthetic 6G RAN telemetry.

Each of these papers covers one or two of our four axes. None benchmarks LSTM, Mamba, and Spiking-SSM jointly under parametric Dirichlet heterogeneity, and none reports per-round NVML energy. Our work is differentiated by the combination of architecture breadth, algorithm breadth, parametric heterogeneity, and energy instrumentation on the **publicly-available** Colosseum/ColO-RAN dataset, in contrast to closed simulators or undisclosed datasets used by Mangi et al.

C. Spiking-SSM and FL $\times$ spiking neural networks

Hybrid spiking-state-space architectures emerged from late 2024. **SpikMamba** [21] combined LIF spiking neurons with Mamba selective scan for event-camera action recognition. **Mamba-Spike** [22] introduced a spiking front-end for general temporal data. **Spiking Point Mamba** [23] extended to 3D point clouds. **SpikingSSMs** [24] formalized sparse-parallel spiking state-space layers. **SpikingMamba** [25] distilled large language models into spiking variants for energy efficiency. **SpikingBrain** [26] scaled spiking architectures to 76B parameters on data-center GPU clusters with explicit FLOP-utilization measurement.

Federated-learning variants of spiking neural networks emerged earlier than spiking-SSM: **arXiv:2106.06579** [27] founded FL $\times$ SNN. **FedLEC** [28] extended to label-skew non-IID. **SNN in vertical FL** [29] examined VFL energy trade-offs. **Robustness of SNN in FL with compression** [30] addressed Byzantine attacks. **Privacy in FL with SNN** [31] characterized gradient leakage. Most recently, **arXiv:2602.12009** [32] examined firing-rate sensitivity to differential privacy in federated SNN — preempting the DP-SNN-FL angle. We accordingly cite this work in Section **??** as a deferral and do not duplicate the DP study; our contribution is orthogonal in its multi-architecture $\times$ parametric-Dirichlet $\times$ NVML scope.

To our knowledge, no published work combines spiking-SSM with FL on cellular/RAN telemetry. Our `spiking_expand2` variant adapts the wide-inner-state design pattern from Mamba [33] to spiking-SSM and demonstrates competitive AUC/energy trade-off on commodity edge GPU under FL.

D. GPU energy measurement methodology

The argument that FLOP counts mis-estimate actual hardware energy predates our work. **Shen et al. 2023** [34] ("Is Conventional SNN Really Efficient?") critiqued synaptic-operation accounting via a Bit Budget framework, finding that quantized ANNs dominate SNNs at equivalent bit budgets. **α-FLOPs** [35] proposed a parallelism-aware alternative. **NeuroBench** [36] standardized neuromorphic benchmarking on real hardware. The **ML.ENERGY Benchmark** [37] measured inference energy on generative AI; **Where Do the Joules Go?** [38] diagnosed the breakdown across model families. **STEP** [6] specifically benchmarked spiking transformers including memory-access cost. **Beyond Backpropagation** [39] applied NVML and CodeCarbon to alternative training objectives.

We follow the **Energy API** approach (`nvmlDeviceGetTotalEnergyConsumption`, available on Volta+ with NVML $\geq$ 11.0) preferred over polling-power Riemann-sum integration per the ML.ENERGY 2025 paper [37] and accompanying leaderboard. Our novel angle is the **federated-training regime**: prior NVML benchmarks measure inference [36], [37] or centralized training [26], [39]; we measure per-round energy across an FL sweep, including idle-baseline subtraction.

E. Heterogeneity in federated learning

NIID-Bench [40] established Dirichlet partitioning as the standard non-IID benchmark. **FedBN** [13] keeps client-local BatchNorm statistics for feature-skew settings; **arXiv:2508.08479** [8] demonstrated FedBN's superiority on cellular feature-skew. **pFedFDA** [41] specifically targets covariate shift. **Borazjani et al. 2025** [42] argued for embedding-based heterogeneity definitions beyond label-skew.

We use Dirichlet partitioning over slice IDs as the canonical covariate-skew model for O-RAN: each gNB serves a different mix of slice traffic, with mixing entropy controlled by α_{dir}. We adopt $\alpha_{\text{dir}} \in \{0.05, 0.10, 0.50, 1.00, 5.00\}$ to span extreme stress (each gNB dominated by a single slice) through near-IID. We additionally evaluate the natural base-station partition (one gNB $\Rightarrow$ one client, 7 ColO-RAN gNBs in the public release) as a non-parametric heterogeneity reference. We discuss the applicability of embedding-based alternatives in Section **??**.

F. Sharpness-aware federated learning (2024–2026)

A separate strand of recent FL work targets *heterogeneity-induced sharp minima* via per-step learning-rate cycling and server-side weight extrapolation. **FedSAM** [43] adapted Foret et al.'s SAM perturbation to the federated client step. **FedSWA** [11] introduced an intra-round cyclical local LR schedule (eq. 3) combined with a LookAhead-style server EMA $\theta_t = \theta_{t-1} + \alpha(v_t - \theta_{t-1})$ with $\alpha = 1.5$ (eq. 17). **FedSCAM** [12] composed SAM with clustering for further sharpness regularization. The reported gains (FedSWA: $+5.6\%$ over FedAvg on CIFAR-100 with 10% client participation across 1000 rounds) assume large-N low-participation FL with coherent client drift toward sharp minima.

We do not empirically evaluate this family on Colosseum/ColO-RAN (see Section **??**). Two structural mismatches make the absence of comparison defensible on mechanism alone. First, our regime — 7 base-station-natural clients with 71% per-round participation, 100 rounds, RAN slice SLA prediction — is *aggregation-noise-dominated* rather than coherent-drift-dominated. Second, FedAdam's adaptive variance damping $\mathrm{Adam}(m, v)$ over $(v_t - \theta_{t-1})$ systematically dominates FedSWA's variance-blind LookAhead overshoot *in expectation* in this regime: FedAdam knows when to step small (high v) and when to step normally (low v), while FedSWA's LookAhead extrapolation rate α_{LA} (the FedSWA paper's α-symbol; we write it α_{LA} throughout to disambiguate from the Dirichlet α_{dir} of Section **??** and the FedDyn α_{dyn} of Section **??**; FedSWA's preregistered value is $\alpha_{\text{LA}} = 1.5$) amplifies *both* signal and noise uniformly. Empirically (Section **??**), FedAdam already saturates the headroom available to server-side aggregation modifications at $+0.006$–$+0.016$ Dirichlet-averaged AUC over FedAvg across the 3 architectures (paired-bootstrap CI_{95} in Section I contribution 2); LookAhead overshoot is unlikely to *systematically* exceed this ceiling without the variance-estimation machinery FedAdam has and FedSWA lacks. We additionally note that FedSWA targets *heterogeneity-as-sharpness-wound*, which our inverted-α finding (Section I, Section **??**) indicates does not characterize this dataset.

References

[1] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "ColO-RAN: Developing machine learning-based xApps for Open RAN closed-loop control on programmable experimental platforms," *IEEE Transactions on Mobile Computing*, 2022.

[2] S. Caldas, S. M. K. Duddu, P. Wu, T. Li, J. Konečný, H. B. McMahan, V. Smith, and A. Talwalkar, "LEAF: A benchmark for federated settings," *arXiv:1812.01097*, 2018.

[3] F. Lai, Y. Dai, X. Zhu, H. V. Madhyastha, and M. Chowdhury, "FedScale: Benchmarking model and system performance of federated learning at scale," in *ICML*, 2022.

[4] o. Song, "FLAIR: Federated learning annotated image repository," 2022.

[5] F. Granqvist *et al.*, "pfl-research: Simulation framework for accelerating federated learning research," in *NeurIPS Datasets and Benchmarks*, 2024, arXiv:2404.06430.

[6] o. Shen, "STEP: A unified spiking transformer evaluation platform for fair and reproducible benchmarking," *arXiv:2505.11151*, 2025.

[7] (authors), "Fed-LSTM: LSTM forecaster for slice-level traffic in cellular FL," 2022.

[8] ——, "FedAvg/FedProx/FedBN benchmark on five throughput-prediction datasets," 2025, arXiv:2508.08479.

[9] Mangi *et al.*, "When Clients Drift: Regime-aware aggregation in 6G RAN telemetry," 2026.

[10] L. Bonati *et al.*, "Colosseum: Large-scale wireless experimentation through hardware-in-the-loop network emulation," *IEEE Open Journal of the Communications Society*, Aug. 2024.

[11] J. Liu *et al.*, "FedSWA: Federated stochastic weight averaging," in *ICML*, 2025, arXiv:2507.20016.

[12] "FedSCAM: Sharpness-aware clustering for federated learning," 2026, arXiv:2601.00853 (Jan 2026).

[13] X. Li, M. Jiang, X. Zhang, M. Kamp, and Q. Dou, "FedBN: Federated learning on non-IID features via local batch normalization," in *ICLR*, 2021, arXiv:2102.07623.

[14] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, T. Parcollet, P. P. B. de Gusmão, and N. D. Lane, "Flower: A friendly federated learning research framework," 2020, arXiv:2007.14390.

[15] (authors), "fev-bench: Forecast evaluation with bootstrap-ci for time-series forecasting benchmarks," 2025.

[16] ——, "Statistical federated learning for beyond-5g SLA-constrained RAN slicing," *IEEE*, 2021.

[17] ——, "CDF-aware federated learning for RAN slicing under long-term SLA constraints," *IEEE*, 2021.

[18] J. Groen *et al.*, "TRACTOR: Traffic analysis and classification tool for open RAN," in *IEEE*, 2023.

[19] M. Barker *et al.*, "REAL: An open-source reinforcement learning framework for the near-rt RIC," 2025, arXiv:2502.00715.

[20] Hayek *et al.*, "Empirical evaluation of federated learning convergence over heterogeneous COTS networks," 2025.

[21] Chen *et al.*, "SpikMamba: When SNN meets Mamba for event-based action recognition," in *(venue TBD)*, 2024.

[22] (authors), "Mamba-Spike: Hybrid spiking front-end with selective state-space backbone for temporal data," 2024, arXiv:2408.11823.

[23] ——, "Spiking point Mamba for 3D point cloud processing," in *ICCV*, 2025.

[24] ——, "SpikingSSMs: Sparse-parallel spiking state-space models," in *AAAI*, 2025.

[25] ——, "SpikingMamba: Distilling large language models into spiking-SSM variants for energy-efficient inference," 2025, arXiv:2510.04595.

[26] ——, "SpikingBrain: Scaling spiking architectures to 76B parameters with explicit FLOP-utilization measurement," 2025, arXiv:2509.05276.

[27] ——, "Federated learning with spiking neural networks," 2021, arXiv:2106.06579.

[28] ——, "FedLEC: Federated learning with spiking neural networks under label-skew non-iid," 2024, arXiv:2412.17305.

[29] ——, "Spiking neural networks in vertical federated learning," 2024, arXiv:2407.17672.

[30] ——, "Robustness of spiking neural networks in federated learning with compression," 2025, arXiv:2501.03306.

[31] ——, “Privacy in federated learning with spiking neural networks,” 2025, arXiv:2511.21181.
[32] ——, “Firing-rate sensitivity to differential privacy in federated spiking neural networks,” 2026, arXiv:2602.12009 (Feb 2026).
[33] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *COLM*, 2024, arXiv:2312.00752.
[34] Shen *et al.*, “Is conventional SNN really efficient? a perspective from bit budget,” *(venue TBD)*, 2023.
[35] A. Asperti *et al.*, “α-FLOPs: A parallelism-aware alternative to flop counting for energy estimation,” *(venue TBD)*, 2021.
[36] J. Yik *et al.*, “NeuroBench: A framework for benchmarking neuromorphic computing algorithms and systems,” *Nature Communications*, 2025.
[37] J.-W. Chung *et al.*, “The ML.ENERGY benchmark: Measuring inference energy of generative AI models,” 2025, arXiv:2505.06371.
[38] ——, “Where do the joules go? diagnosing energy breakdowns across generative-model families,” 2026.
[39] Spyra *et al.*, “Beyond backpropagation: NVML-instrumented energy profiling of alternative training objectives,” 2025.
[40] Q. Li, Y. Diao, Q. Chen, and B. He, “NIID-Bench: Federated learning on non-IID data silos: An experimental study,” in *ICDE*, 2022.
[41] (authors), “pFedFDA: Personalized federated learning under covariate shift via feature-distribution alignment,” in *NeurIPS*, 2024.
[42] Borazjani *et al.*, “Embedding-based heterogeneity definitions for federated learning beyond label-skew,” 2025, arXiv:2503.14553.
[43] D. Caldarola, B. Caputo, and M. Ciccone, “Improving generalization in federated learning by seeking flat minima (FedSAM),” in *ECCV*, 2022.